



Road_Accident_Severity_Analyzer

Data Science and Artificial Intelligence

Komal Kumari

Pallavi

Shruti Saini

Road Accident Severity and Safety Hotspot Analyzer

Project Report

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Submitted By

- Komal Kumari
 - Pallavi
 - Shruti Saini
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Certificate

- This is to certify that this project titled "**Road Accident Severity and Safety Hotspot Analyzer**" was carried out as part of the Himshikhar 2026 Project under the Data Science & Artificial Intelligence Track.
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 - I would also like to acknowledge the contributors of the Kaggle Road Accident Severity Dataset and the open-source Python community.
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Chapter 1 — Abstract

- Road accidents are a significant public safety issue worldwide. Predicting accident severity and identifying accident-prone conditions can assist authorities in implementing effective preventive measures.
 - This project develops a machine learning-based system capable of predicting accident severity while identifying high-risk patterns using accident-related features such as weather conditions, road surface conditions, driver characteristics, and environmental factors.
 - The project also provides safety recommendations to assist policymakers and transportation authorities in improving road safety.
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Chapter 2 — Introduction

- Road traffic accidents result in millions of injuries and fatalities every year.
 - Traditional accident analysis mainly relies on descriptive statistics.
 - Machine Learning enables:
 - Pattern recognition
 - Severity prediction
 - Risk analysis
 - Better decision-making
 - This project combines predictive analytics with hotspot analysis to support safer transportation planning.
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Chapter 3 — Problem Statement

- The objective of this project is to develop a predictive system capable of:
 - Predicting accident severity.
 - Identifying high-risk accident conditions.
 - Explaining important contributing factors.
 - Providing actionable safety recommendations.
 - The solution should assist traffic authorities in planning preventive strategies rather than only displaying predictions.
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Chapter 4 — Objectives

- The major objectives are:
 - Study road accident data.
 - Clean and preprocess the dataset.
 - Perform exploratory data analysis.
 - Train machine learning models.
 - Compare model performance.
 - Identify accident hotspots.
 - Provide practical safety recommendations.
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Chapter 5 — Literature Review

- Discuss previous work on:
 - Road accident prediction.

- Machine learning in transportation.
 - Decision Trees.
 - Random Forest.
 - Classification techniques.
 - Mention that previous research has shown machine learning can significantly improve accident severity prediction.
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Chapter 6 — Dataset Description

- Dataset Name: Road Accident Severity Dataset
 - Source: Kaggle
 - Number of Records: 12,316
 - Number of Features: 32
 - Target Variable: Accident Severity
 - Describe important features such as: Weather Conditions, Driver Age, Driver Experience, Vehicle Type, Road Surface Conditions, Light Conditions, Cause of Accident
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Chapter 7 — Methodology

- Include a workflow diagram.

Problem Understanding

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Dataset Collection

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Data Cleaning

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EDA

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Feature Engineering

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Machine Learning

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Evaluation

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Hotspot Analys

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Safety Recommendations

Chapter 8 — Data Preprocessing

- Explain:
 - Missing value handling
 - Duplicate removal
 - Data type conversion
 - Encoding categorical variables
 - Train-test split
- Include code snippets only if required; otherwise explain the process in words.

Chapter 9 — Exploratory Data Analysis

- Insert your graphs: Severity Distribution, Driver Age, Weather Conditions, Light Conditions, Road Surface, Driver Experience, Cause of Accident
- Below every graph write observations.
- Observation:
 1. Slight injuries constitute the largest proportion of accidents.
 2. Rainy conditions contribute to more severe accidents.
 3. Young drivers are involved in a larger number of accidents.

Chapter 10 — Feature Engineering

- Explain:
 1. Label Encoding
 2. Feature Selection
 3. Target Variable
 4. Train-Test Split
- Include a table:

Step	Description
Missing Value Handling	Filled using mode/median
Encoding	LabelEncoder
Train-Test Split	80-20

Chapter 11 — Machine Learning Models

- Describe:

Logistic Regression

- Advantages
- Limitations

Decision Tree

- Advantages
- Limitations

Random Forest

- Advantages
- Why selected as final model

Chapter 12 — Model Evaluation

- Include:
 1. Accuracy
 2. Precision
 3. Recall
 4. F1 Score
 5. Confusion Matrix
 6. Classification Report
 7. Feature Importance
 8. Add screenshots of:
 9. Confusion Matrix
 10. Feature Importance
 11. Explain what each metric means in simple language.

Chapter 13 — Safety Hotspot Analysis

- Discuss:
 1. High-risk weather conditions
 2. High-risk road surfaces
 3. Driver age groups
 4. Common accident causes
 5. Lighting conditions
- Provide 4-6 key findings based on your analysis.

Chapter 14 — Results and Discussion

- Discuss:
 1. Best model
 2. Prediction performance
 3. Key accident factors
 4. Overall project performance
- Example: Random Forest achieved the highest prediction accuracy among the evaluated models and effectively classified accident severity. Weather conditions, road surface conditions, and driver-related factors were identified as major contributors to accident severity.

Chapter 15 — Limitations

- Mention:
 1. Historical dataset only
 2. No live traffic data
 3. No GPS-based hotspot mapping
 4. Limited geographic scope
 5. Dependent on dataset quality
 6. Predictions should support, not replace, expert judgment
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Chapter 17 — Future Scope

- Possible improvements:
 1. Real-time prediction
 2. Google Maps integration
 3. Weather API
 4. Deep Learning
 5. XGBoost
 6. Cloud deployment
 7. Mobile application
 8. Interactive dashboards
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Chapter 18 — Conclusion

- Summarize:
 1. The project successfully developed a machine learning-based system capable of predicting accident severity and identifying high-risk accident patterns.
 2. The integration of predictive analytics and hotspot analysis provides practical support for road safety planning and demonstrates the usefulness of AI in transportation safety.
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References

- Include references such as:
 1. Kaggle Road Accident Severity Dataset
 2. Scikit-learn Documentation
 3. Pandas Documentation
 4. Streamlit Documentation
 5. Matplotlib Documentation
 6. Seaborn Documentation
 7. Relevant research papers on accident severity prediction
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Appendix

- Include:
 1. Folder Structure
 2. Sample Code
 3. Model Comparison Table
 4. Additional Graphs
 5. Screenshots